Gauge transformation showcase: validation report

Abstract

The question is whether two visibly different vector potentials represent the same uniform magnetic field. The showcase uses $A=(0,bx,0)$ and $A'=(-by,0,0)$. Their difference is the gradient of $s=-bxy+C$; both curls equal $(0,0,b)$. An interpolating family $A\lambda=A+\lambda\nabla s$ preserves this curl and all closed-loop circulation. Exact differentiation establishes the result; numerical experiments and browser checks test whether the implementation represents it faithfully.

Methodology and evidence

We checked the reference passages, derived the connecting scalar, tested the mathematics before implementation, and reused one sampled grid in both renderers. The experiments below document the hypothesis, evidence, problems, and corrections without treating finite numerical samples as a proof.

- Experiment 1: exploratory central-difference check.
- Experiment 2: test-driven physics and scientific exports.
- Experiment 3: native slider interaction and responsive rendering.
- Experiment 4: factual audit and showcase presentation.
- Experiment 5: independent recheck of physical assumptions and references.
- Acceptance and review: completed checks, corrections, and explicit limits.

The PDF companion assembles this summary, the individual experiment records, and the generated four-panel figure from their existing sources. It is a local reproducibility report, not a claim of new experimental physics.

Conclusion and limitations

Gauge freedom changes the vector potential while preserving the magnetic field. Here b is constant and the gauge function is time-independent and smooth over all space. Coordinates and magnitudes use illustrative units. The plotted slice contains all spatial dependence because the fields have no z dependence. The browser checks cover Chromium; the interactive renderer needs access to its pinned D3 CDN. The scientific PNG/PDF and the algebra can be read offline.

The reference passages provide the conceptual starting point. Independent corroboration is MIT's electromagnetism gauge and circulation derivation, page 6.

Experiment 1 — independently check the proposed gauge family

Date: 2026-10-10 (Europe/Berlin).

Question and hypothesis

Does $A_\lambda=(-\lambda by,(1-\lambda)bx,0)$ have curl $(0,0,b)$ for every selected gauge? Differentiation predicts $(1-\lambda)b+\lambda b=b$. The corresponding gauge function is $s=-bxy$, and its constant offset is irrelevant.

Method

During planning, an in-memory Python calculation evaluated central differences with step 0.0001 at (-2,1), (0,0), and (1,-2). It combined $b=-1,0,1$ with $\lambda=0,0.25,0.5,0.75,1$, giving 45 point/gauge/strength cases. The test compared the measured z curl with b using absolute tolerance 1e-9. The implementation session's tool output is the evidence for this exploratory run; it did not create a raw log file or alter the repository.

Result and interpretation

All 45 cases passed. The maximum absolute error was approximately 1.56e-12. This small residual is floating-point subtraction error in the finite difference, not a physical change in the magnetic field. The analytic result remains exact. Subsequent grid tests use exactly representable sample spacing and are documented separately rather than retroactively replacing this result.

Learning and choice

The continuous family gives the slider a direct mathematical interpretation. Its midpoint is the symmetric gauge, making rotational symmetry visible. We chose this family because it connects the two potentials directly without adding another physical model.

The mathematical convention and invariant closed-loop integral were checked against MIT's solution, page 6.

Experiment 2 — test the field model and inspect scientific figures

Date: 2026-10-10 (Europe/Berlin).

Question and hypothesis

Can one small implementation produce comparable plots without losing the model's field signs, gradient relation, or circulation? Shared coordinates, equal aspect, and a fixed arrow scale should preserve meaningful comparisons.

Method and evidence

We wrote endpoint, midpoint, scalar-gradient, curl, and circulation tests before implementing the model. The first test run failed during collection because the field module did not yet exist. After implementation, 22 tests passed. The production validation evaluates 15 gauge/strength combinations on a 17-by-17 grid and checks a counterclockwise unit square independently.

The first complete command finished at 11:55:41 UTC. Its timestamped generation log recorded zero maximum residuals for the grid curl, gradient identity, and unit-square circulation, with a tolerance of 1e-10. These affine/bilinear fields and dyadic sample spacing yield exact results in those particular floating-point calculations; this does not imply arbitrary numerical derivatives are exact. Each run saves current residuals in outputs/validation.json.

Issue, correction, and result

Visual inspection found small diagonal artifacts in the zero scalar contour near the saddle at the origin. The exact zero contour is the pair of coordinate axes. We therefore retained the exact axes and plotted only nonzero contour levels. This removes an ambiguous contour interpolation at a degenerate crossing without changing field data. The corrected pipeline completed at 11:56:14 UTC.

The PNG was inspected for arrows, sign changes, labels, contour geometry, and uniform perpendicular B symbols. The PDF is checked for one page and no raster images. The working mathematical core and its outputs were pushed in commit 84f11f8 before the interactive feature and final review.

Learning and choice

Matplotlib was sufficient; a custom scientific renderer would add unnecessary work. Its quiver documentation guided Cartesian arrow directions, explicit scale, and common pivot. Zero vectors are omitted. The separate B panel avoids projecting the entire magnetic field to zero in the xy plane.

Experiment 3 — verify interaction and responsive presentation

Date: 2026-10-10 (Europe/Berlin).

Question and hypothesis

Does moving the gauge slider visibly change A while leaving B unchanged, and does the presentation remain readable on narrow screens? A native range input and fixed sampled endpoints should support this without a backend.

Method

The generator embeds its existing endpoint samples in the browser view. Chromium opens the generated local file, uses End and Home keys to reach the two endpoint potentials, and applies 50 ArrowRight presses to reach λ=0.50. The check compares all potential arrow paths and the complete B-symbol group. It then inspects equal aspect and horizontal overflow at widths 736, 360, and 320 pixels. Runs cover b=-1,0,1; zero field is expected to remain empty. Browser captures and timestamped console logs are kept locally in ignored evidence directories.

Issues and corrections

The in-app browser connection rejected the Linux workspace before opening a page, so verification uses a separate local Playwright Chromium session. It does not attach to a personal browser. The first integrated run passed 25 tests but failed three responsive checks because SVGRect's width/height getters were not serialized by the test harness. All slider and B-invariance assertions had already passed. Explicitly returning plain width/height properties repairs the measurement, following MDN's SVGRect documentation and Playwright's evaluation contract.

Result and learning

The browser-only rerun passed all three field-sign cases in 12.53 seconds; the subsequent full run passed 37 tests. The acceptance record lists the inspected evidence. The failure showed why a test harness needs inspection as well as the product. Independent endpoint data checks and native keyboard interaction complement the geometric measurements. We keep the browser's field formulas out of the renderer: interpolation reuses the Python-generated samples.

The only runtime network dependency is the pinned D3 script. If it cannot load, the page presents an accessible error and disables its slider instead of silently displaying an empty successful-looking chart.

Experiment 4 — factual audit of the gauge transformation showcase

Date: 2026-10-10 (Europe/Berlin).

Question and method

Does the explanation accurately connect changing vector-potential arrows to an invariant magnetic field? We compared the reference passages with the written derivation, differentiated the potentials and connecting scalar, and checked the circulation argument. MIT's electromagnetic derivation, page 6 independently supports gradient gauge changes, invariant fields, and unchanged closed-loop circulation. Existing export and browser checks were updated before renaming the generated files. The first export run failed at the expected new PNG path, confirming that the tests required the new deliverable names.

Findings and corrections

The core mathematics was correct: both curls are (0,0,b), their difference is the gradient of s=-bxy+C, and the slider midpoint is one half of B cross r. No field formula needed to change. We clarified the explanation in four places:

- Curl's z component is oriented circulation per unit area in the shrinking-loop limit. Each nonzero square-side contribution is $bL^2/2$ for the displayed square centered at the origin; an off-center square preserves the total bL^2.

- The zero contour is the two axes only when b is nonzero and C=0. At b=0, s=0 everywhere for C=0. The signed component b has field magnitude |b|.

- The slider selects spatially constant, static gauges. It does not represent physical time. The electric-potential transformation uses an explicitly stated SI convention; the gauge function is distinct from electric potential.

- The gradient identity uses smooth, single-valued functions. Their explicit connecting scalar establishes the relation globally for this example.

Presentation and reasoning

Gauge transformation is now the primary subject in titles, plot labels, browser metadata, package metadata, and artifact names. Selected reference passages retain their wording and remain separate from the added explanation. Broad historical and formalism claims are outside this static classical demonstration and are omitted. No unverified book attribution is introduced.

We retained the existing NumPy, Matplotlib, and sampled-data browser design: the review found an explanation and presentation issue, not a new mathematical model or rendering requirement. The PNG and vector PDF use the same shared arrow scale as before. The report is still generated from these Markdown records.

Validation and review

uv run gauge-fields completed at 12:30:27 UTC. All 15 gauge/strength cases had zero measured curl, gradient, and circulation residuals at tolerance 1e-10. uv run pytest -q passed all 37 tests in 16.41 seconds, including complete generation for b=-1,0,1 and native keyboard/browser checks at 736, 360, and 320 pixels. Seven new generation/browser logs contained no warning or error entries. The figure, desktop/phone views, and report page were visually inspected; the one-page figure PDF contains no raster images.

One review pass checked factual scope, reference fidelity, artifact paths, documentation, and unchanged numeric-only browser data. GitHub reported the repository's current canonical name as magnetic-field-gauge-transformation; clone instructions and PDF links now use it. No dependencies or mathematical formulas changed. Browser-engine and host-integration limits remain documented in the initial acceptance record.

Experiment 5 — independently recheck the explanation

Date: 2026-10-10 (Europe/Berlin).

Question and method

Does the current explanation contain incorrect equations, unsupported physical claims, or invented source attributions? We checked each mathematical step against the implemented field functions and independently differentiated the potentials and connecting scalar. We inspected the square's directed edge contributions, parameter assumptions, dimensional units, and the relation between the stated arrow scale and both renderers.

MIT's derivation, page 6 confirms the electromagnetic gauge transformation and invariant closed-loop integral. The official-domain indexed excerpt of David Tong's Cambridge notes, section 1.2.2, equation (1.18) independently identifies the midpoint potential as the symmetric gauge. Direct PDF retrieval was unavailable during this check; the excerpt, not a full reading of that document, is the corroborating evidence. Only that mathematical definition is used; the showcase remains classical.

Findings and changes

The core derivation was already correct. No field formula or visualization needed to change. Three statements were made more precise:

- The zero-field scalar retains its arbitrary additive constant; zero everywhere specifically refers to the displayed choice of that constant.
- Translating a circulation square can change individual side contributions, but does not necessarily change each of them. The total remains invariant.
- Matching magnetic fields alone does not establish matching electric fields, even for static potentials. The showcase holds the electric scalar potential fixed, which makes its static gauge choices electromagnetically equivalent.

We also defined the position vector used in the midpoint formula and specified the axis of rotational symmetry. Reference quotations remain unchanged; no new book attribution or unsupported physical claim was found in the explanation.

Verification and limits

uv run pytest tests/test_fields.py -q passed all 22 mathematical tests in 0.20 seconds. They cover both field signs and zero, intermediate gauges, independent finite differences, an off-center circulation loop, and the additive scalar constant. These checks support the implementation; direct differentiation establishes the exact identities. This follow-up changes prose and references only, so the prior browser results remain the relevant interface evidence rather than a newly claimed browser run.

The complete generation command succeeded at 12:37:52 UTC with no warnings or errors in its log. Its 15 gauge/strength cases again had zero measured residuals. The updated eight-page PDF includes this audit; the plot and interactive file contents were checked to be unchanged.

Acceptance and review record

Date: 2026-10-10 (Europe/Berlin). This record describes the implemented project and the checks performed in the initial implementation. The subsequent factual audit records the latest validation.

Verified results

- uv run pytest -q: 37 tests passed in 15.84 seconds after the mathematical, rendering, configuration, report, and browser changes. The suite generates complete artifacts independently for b=-1,0,1.

- uv run gauge-fields: successful default generation at 12:05:05 UTC. The generator's 15 gauge/strength cases use 289 grid points each; maximum absolute curl, gradient, and circulation residuals were all zero at tolerance 1e-10.

- Twenty timestamped generation/browser logs inspected at this checkpoint contained no WARNING, ERROR, or browser error entries. Earlier expected test-development failures are documented in the experiment records.

- Scientific PNG and vector PDF inspected; the figure PDF has one page and no raster images. The companion's five initial pages contain selectable experiment text and a figure appendix. This acceptance record adds a sixth page when the companion is regenerated.

- Browser behavior checked through native Home, End, and ArrowRight keys. B geometry remains unchanged at the endpoints and midpoint. The zero-field case correctly has no arrows. Layouts at 736, 360, and 320 pixels retain square plotting frames and have no horizontal overflow. Desktop and phone browser captures were inspected, as was the PDF's first rendered page.

One-pass code and documentation review

Correctness: checked the signs in A′−A, coordinate/derivative ordering, contour behavior at the scalar saddle, and the difference between a gauge parameter and physical time. Reference passages are clearly distinguished from added prose.

Security: numeric-only serialized data is embedded in the visualization; no credentials or user-provided executable markup enter it. The repository is private, and .env, .venv, raw logs, and local browser captures are ignored. External browser loading is limited to the version-pinned D3 script.

Maintainability and design: the thin pipeline delegates to configuration, model, verification, two renderers, and a report exporter. The browser reuses the same endpoint samples; the PDF reads existing Markdown instead of a second copy of report prose. The architecture, setup, settings, and outputs in README were checked against code and observed runs.

Reliability: configuration rejects nonfinite and unusable values, the zero field avoids normalization errors, failures preserve nonzero exit status and logged tracebacks, and a missing CDN yields a visible error state. Exact derivation and independent finite differences/line integrals provide different forms of evidence; they are not presented as interchangeable proofs.

Review corrections and limits

Git initially treated the ReportLab file as text because its initial bytes did not trigger binary detection, producing irrelevant PDF whitespace diagnostics. Explicit PDF/PNG binary attributes fix review and line-ending handling. PDF links to individual experiments now resolve to the private repository, so they remain meaningful when the PDF is downloaded separately. These are small review corrections; the prior physics and browser results still apply.

The interactive view was verified in a separate Chromium instance, not in the unavailable in-app browser connection. Other browser engines and the host's optional saved-state integration were not separately exercised. The default layout is manually inspected; parameter boundary checks do not claim exhaustive visual review of every possible configuration. Runtime dependencies, supported configuration ranges, and the CDN requirement are documented in README.

Generated field comparison

